Dataset with brain atlas A: for the first set of 5 small programming tasks use the Oasis dataset provided by the *nilearn* package, which can be downloaded using the code below. This is a dataset of structural brain scans (aka voxel-based morphometry) for male and female individuals.

```
import numpy as np
from nilearn import datasets
from nilearn.input_data import NiftiLabelsMasker
from nilearn.image import index_img
import nibabel as nib

brain_data = datasets.fetch_oasis_vbm(n_subjects=100)
yeo = datasets.fetch_atlas_schaefer_2018(n_rois=200)
masker = NiftiLabelsMasker(labels_img=yeo.maps, standardize=True,
memory='nilearn_cache')
input_variables =
masker.fit_transform(brain_data.gray_matter_maps)
output_variable = np.array(brain_data.ext_vars.mf == 'F',
dtype=int) #gives 1 for females and 0 for males
```

1. Use `sklearn.linear_model.LogisticRegression`(hyperparameter set to 1.0) to fit separate logistic predictive models based on 100, 200, 300, …, 900, 1000 **brain regions of interest** (see 'n_rois' argument for datasets.fetch_atlas_schaefer_2018) based on the first 80% of the 100 structural brain scans to predict sex differences. Next, evaluate each fitted LogReg model on the unseen final 20% of the 100 structural brain scans and retrieve classification accuracy in XX.YY%. Please plot the classification accuracy on the test set (x axis=the different atlas resolutions, y axis=prediction accuracy on test set).

2. Based on **1000 brain regions of interest** (see 'n_rois') and using the **first 80% of the 100 structural brain scans**, use (`sklearn.decomposition.PCA`) **with n_components=1,2,3,4,…25** to perform **principal component analysis** and then use `sklearn.linear_model.LogisticRegression`(hyperparameter set to 1.0) to fit **separate logistic regression models** to predict sex differences. The input features for fitting the LogReg are only the **component projections**. Next, evaluate each fitted LogReg model on the **unseen final 20% of the 100 structural brain scans** and retrieve classification accuracy in XX.YY%. Please plot the classification accuracy on the test set (x axis=the different numbers of PCA components, y axis=prediction accuracy on test set).

3. Based on **1000 brain regions of interest** (see 'n_rois') and using the **first 80% of the 100 structural brain scans**, use (`sklearn.cluster.KMeans`) to perform k means clustering with n_clusters=1,2,3,4,…25. Then use `sklearn.linear_model.LogisticRegression`(hyperparameter set to 1.0) to fit separate logistic regression models to predict sex differences. So, the input

features for fitting the LogReg are only the **cluster projections.** Next, evaluate each fitted LogReg model on **the unseen final 20% of the 100 structural brain scans** and retrieve classification accuracy in XX.YY%. Please plot the classification accuracy on the test set (x axis=the different numbers of k-means clusters, y axis=prediction accuracy on test set).

4. Based on **1000 brain regions of interest** (see 'n_rois') and using the **first 80% of the 100 structural brain scans**, use (`sklearn.feature_selection.SelectKBest`) and (`sklearn.feature_selection.f_classif`) to perform feature selection with the best k=2,4, 6, 8, 10 features. Then, use `sklearn.linear_model.LogisticRegression`(hyperparameter set to 1.0) to fit separate logistic regression models to predict sex differences. So, the input features for fitting the LogReg are only the k best single features, as identified on the training data. Next, evaluate each fitted LogReg model on the unseen final 20% of the 100 structural brain scans and retrieve classification accuracy in XX.YY%. Please plot the classification accuracy on the test set (x axis=the different numbers of k best features, y axis=prediction accuracy on test set).

5. Use `sklearn.linear_model.LogisticRegression`(hyperparameter set to 1.0) to fit separate logistic predictive models based on 100, 200, 300, …, 900, 1000 brain regions of interest (see 'n_rois') based on the first 80% of the 100 structural brain scans to predict sex differences. Use the probabilistic prediction of LogReg (`predict_proba` function) from the training data from the 10 already obtained LogReg models to **fit a LogReg stacking model** to predict sex. Next, *evaluate the final LogReg stacking model* on the unseen final 20% of the 100 structural brain scans and print classification accuracy in XX.YY%.

Dataset with brain atlas B: For the second set of 5 small programming tasks use the Oasis dataset provided by the *nilearn* package, which can be downloaded using the code below. This is a dataset of structural brain scans (aka voxel-based morphometry) for male and female individuals.

```
import numpy as np
from nilearn import datasets
from nilearn.input_data import NiftiLabelsMasker
from nilearn.image import index_img
import nibabel as nib

brain_data =
datasets.fetch_oasis_vbm(n_subjects=100)

crad = datasets.fetch_atlas_craddock_2012()
atlas_nii = index_img(crad['scorr_mean'], 11) #this needs to be
```

```
modified for some tasks
masker = NiftiLabelsMasker(labels_img= atlas_nii, standardize=True)
input_variables =
masker.fit_transform(brain_data.gray_matter_maps)
output_variable = np.array(brain_data.ext_vars.mf == 'F',
dtype=int) #gives 1 for females and 0 for males
```

6. analogous to task 1, but using 43 different versions of Craddock region atlas resolutions (.scorr_mean) instead of the 10 region numbers from Dataset A.

7. analogous to task 2, but using version 43 of Craddock region atlas with highest resolutions (.scorr_mean).

8. analogous to task 3, but using version 43 of Craddock region atlas with highest resolutions (.scorr_mean).

9. analogous to task 4, but using version 43 of Craddock region atlas with highest resolutions (.scorr_mean).

10. analogous to task 5, but using the 43 different Craddock region atlas resolutions (.scorr_mean) instead of the 10 region numbers from Dataset A.